MODULE 36 · EXAM STRATEGY - ALL SECTIONS

Confusion pairs

Every tiebreaker from the course side by side, then eight scenarios that combine them.

SECTION All sections EXAM WEIGHT - MODULE 36 of 40

admin@sthithapragna.com

Most wrong options are **a true statement about the wrong thing**. The skill is noticing which question each option actually answers.

Pairs from sections 1 and 2

PAIR	DECIDED BY
Chain or agent	Could you draw the flowchart before the request?
Router or agent	Does one handler complete each request type?
Retrieval or fine-tuning	Knowledge problem, or behaviour problem?
Classify or extract	Choose from a known list, or copy out of the text?
Keyed lookup or retrieval	Is there an ID for one entity, or general prose?
Tool or retrieval	A value in a system, or a passage in a document?
Remove at ingestion or filter at query	Should nobody see it, or only some queries?

ON THE EXAM

In design questions, name the task first. The component follows from it.

Pairs from sections 2 and 4

PAIR	DECIDED BY
Small or large chunks	Precision lost, or context lost?
Parent-child or bigger chunks	Search small, return big, when both failed
Embedding window or LLM window	One chunk, or k chunks plus prompt and answer?
Recall or precision	Never found, or found but ranked low?
Reranker on or off	Quality first with seconds to spare, or sub-100 ms?
Standard or storage-optimised	Latency critical under 320M, or scale and cost?
Triggered or continuous sync	Scheduled data, or seconds of freshness?

ON THE EXAM

In retrieval questions, find which stage failed before choosing a fix: source, chunking, recall or ranking.

Pairs from sections 1 and 3

PAIR	DECIDED BY
Bigger or smaller model	Smallest that passes your evaluation
Reasoning or not	Multi-step problem, or fast simple answer?
Base or instruct	Fine-tuning, or serving?
Asking or enforcing a format	Does a program read it, and did asking fail?
Positive or negative instruction	Describe the target; keep bans for hard limits
System prompt or guardrail	Behaviour, or a rule that must be enforced?
Block or redact	Is the content forbidden, or must the task continue?

ON THE EXAM

In model and prompt questions, apply hard constraints first, then the cheapest option that passes.

Pairs from sections 1, 3 and 4

PAIR	DECIDED BY
LangChain or LangGraph	A line, or loops, state and approvals?
Knowledge Assistant or custom rag	Cited Q and A, or a stated need for control?
Supervisor Agent or custom orchestrator	Coordinating existing agents, or routing that needs code?
Extraction or an assistant	Same fields from every document, or a new question each time?
Genie or retrieval	A number from rows, or a passage from a document?
Delta Sync or Direct Vector Access	Content in a table, or vectors your code writes?
Stages or aliases	Unity Catalog uses aliases

ON THE EXAM

Walk the builders before writing code. Databricks' own advice, and the exam's default, is start simple.

Pairs from section 4

PAIR	DECIDED BY
Pay-per-token or provisioned	Exploring and spiky, or guarantees and fine-tuned weights?
Priority or provisioned	Consistent latency with no commitment, or reserved capacity?
Endpoint or ai_query batch	Is anyone waiting, or is it a table?
Automatic passthrough or on-behalf-of-user	Same data for all, or per-user access and audit?
Genie app or custom app	Data questions in chat tools, or your own agent and UI?
Chatting in Slack or posting to Slack	Users ask, or the agent notifies?
Short-term or long-term memory	Forgets within a conversation, or between them?

ON THE EXAM

In security questions, security lives in Unity Catalog and identity, never in a prompt.

Pairs from sections 5 and 6

PAIR	DECIDED BY
Direct or indirect injection	From the user, or hidden in content?
Publicly available or licensed	Only the terms decide
Evaluation or monitoring	Is the change better, or how is the live app doing?
Judges with or without expectations	Do you have the mark scheme?
LLM judge or code scorer	Needs judgement, or checkable exactly?
Average or p95 latency	Users feel the tail
Budget alert or rate limit	Tell you afterwards, or stop it now?

ON THE EXAM

In evaluation questions, check what the judge needs, and whether the data or traffic can supply it.

> A telecom assistant must answer: why is my bill higher this month? Signed-in customers' bill lines sit in a table keyed by account number.
>
> The tariff rules that explain price changes are in a PDF document.

Which design fits best?

A Index both the bill tables and the tariff PDF for retrieval

B Look up bill lines by account, and retrieve tariff passages

C Fine-tune a model on past bills and on the tariff text

D Ask a Genie Agent to read the tariff PDF as well as bills

◆ PAUSE AND DECIDE

Which design fits best?

A Index everything

Bill values become fuzzy, stale text.

B Keyed lookup plus retrieval

Values by ID from the table, passages from the PDF.

C Fine-tune

Knowledge problem, and bills change monthly.

D Genie reads the PDF

Genie works over tables only.

THE SIGNAL

Two pairs at once: keyed lookup or retrieval, and Genie or documents. Each kind of data gets its own home.

> After moving from BGE to GTE embeddings, a team raised its chunk size to 4,000 tokens. It sends the top 8 chunks to an LLM with a 16,384-token window.
>
> Requests now fail with a maximum context length error.

What should the team change?

A Switch back to the 512-token BGE embedding model

B Reduce the chunk size or k, so the prompt fits the LLM

C Increase the embedding dimensions to 1,024 or more

D Enable the reranker so the chunks are better ordered

◆ PAUSE AND DECIDE

What should the team change?

A Back to BGE

Would then truncate 4,000-token chunks. Wrong limit.

B Smaller chunks or fewer

8 times 4,000 is 32,000: shrink k or the chunk.

C More dimensions

Vector size has nothing to do with prompt length.

D Reranker

Reorders the same 32,000 tokens.

THE SIGNAL

Maximum context length is the LLM's window. GTE made big chunks possible; the LLM cannot hold eight of them.

> A nightly ai_query job extracts fields from 3 million support emails into typed columns.
>
> The prompt already lists the fields and includes an example, but about 1% of outputs still fail to parse.

What is the most reliable fix?

A Pass a JSON schema to ai_query through responseFormat

B Rewrite the prompt in capitals, and repeat it twice

C Move the whole job to priority pay-per-token instead

D Split the table into many smaller batches of rows

◆ PAUSE AND DECIDE

admin@sthithapragna.com

What is the most reliable fix?

A responseFormat schema

Enforces the shape. Asking already failed.

B Capitals, repeated

Still asking, and the prompt was already good.

C Priority mode

Interactive speed at a higher rate. Not a format fix.

D Smaller batches

Reduces throughput. Not a format fix.

THE SIGNAL

Asking or enforcing, and batch best practice. The prompt was already good, so move from asking to enforcing.

> A Databricks App shows HR data to managers. A Unity Catalog row filter already limits each manager to their own team's rows.
>
> The app must not reimplement the filtering.

How should the app authenticate to the data?

A App authorisation, with filtering done in the Python code

B Automatic passthrough, using the deployer's own identity

C User authorisation, forwarding each manager's own token

D A shared service principal with access to every row

◆ PAUSE AND DECIDE

How should the app authenticate to the data?

A App auth, filter in Python

Reimplements the filter, which is ruled out.

B Deployer's identity

Everyone sees what the deployer sees.

C User authorisation

Each manager's token, so the row filter applies.

D Shared service principal

One identity sees every row.

THE SIGNAL

Per-user data and an existing row filter. Only the user's own identity lets Unity Catalog apply it.

> A team wants automatic quality checks on its live assistant's traffic, every day, without writing expected answers.
>
> It already runs a test set before each release.

What should it set up?

A. Register the Correctness judge to run on all production traces

B. Run mlflow.genai.evaluate on last month's test set every day

C. Ask every user to rate each answer with a thumbs-up button

D. Register Guidelines and Safety scorers with a sample rate

◆ PAUSE AND DECIDE

What should it set up?

A Correctness on production

Needs expected facts, which live traffic lacks.

B Daily evaluation on a test set

Evaluation, not monitoring. Not live traffic.

C Thumbs on every answer

Not automatic, and sparse and noisy.

D Guidelines and Safety, sampled

No expectations needed. Runs on live traces.

THE SIGNAL

Live traffic and no expected answers: two pairs at once, monitoring over evaluation, and judges without a mark scheme.

> A translation assistant sends customer emails to an externally hosted model through Unity AI Gateway.
>
> Postal addresses must never reach the provider, but every email must still be translated.

Which guardrail fits?

A PII blocking on the input phase of the gateway

B PII redaction on the input phase of the gateway

C PII redaction on the output phase of the gateway

D A jailbreak guardrail on the gateway's input phase

◆ PAUSE AND DECIDE

Which guardrail fits?

A Blocking on input

Stops the translation, which must still happen.

B Redaction on input

Addresses masked before the call; translation continues.

C Redaction on output

Too late: the address already reached the provider.

D Jailbreak guardrail

Wrong threat entirely.

THE SIGNAL

Block or redact, and input or output. Must never reach means input; must still be translated means redact.

> An internal analytics tool searches 400 million embedded research notes. About 20 queries per second arrive, and a second per query is acceptable.
>
> The notes are refreshed from a nightly load. Cost matters.

Which AI Search configuration fits?

A. A standard endpoint with a high target QPS set

B. A standard endpoint using continuous index sync

C. A storage-optimised endpoint with triggered sync

D. Several standard endpoints, one for each region

◆ PAUSE AND DECIDE

Which AI Search configuration fits?

A Standard, high target QPS

400M is beyond standard capacity; QPS is modest.

B Standard, continuous

Over capacity, and pays for freshness nobody needs.

C Storage-optimised, triggered

Holds 1B, cheaper, latency fine, nightly sync.

D Several standard endpoints

Costly workaround, and splitting by region is arbitrary.

THE SIGNAL

Standard or storage-optimised, and triggered or continuous. Size, latency, cost and a nightly load all point one way.

admin@sthithapragna.com

> Finance wants to be warned when month-to-date model serving spend passes set amounts.
>
> It does not want requests blocked, only an early warning.

What should the platform team set up?

A. A budget with alert thresholds on the serving spend

B. Per-user rate limits configured on every endpoint

C. A daily query of inference tables for payload size

D. Guardrails in Log mode on every serving endpoint

◆ PAUSE AND DECIDE

What should the platform team set up?

A Budget with alerts

Warns at thresholds, blocks nothing.

B Rate limits

Blocks requests, which finance does not want.

C Payload size query

Payload size is not spend.

D Guardrails in Log mode

Records policy checks, not cost.

THE SIGNAL

Budget alert or rate limit. Warn, do not block: a budget. It is the one case where the budget is the right answer.

What to carry forward

- Most wrong options are **true statements about the wrong thing**.
- For each option, ask which question it answers.
- Harder questions combine **two pairs**. Resolve each, and the answer is usually forced.
- Revise the six pair tables in the last days before the exam.

> ON THE EXAM
>
> Next: module 37 shows how Databricks writes these questions, and how to read them under time pressure.